Renewable & Non-Renewable Energy

OBJECTIVES

In this lesson, students will develop an understanding of renewable and non-renewable energy sources. They will conduct their own research and present their findings to the class.

ACARA CONTENT DESCRIPTIONS

Some of Earth's resources are renewable, including water that cycles through the environment, but others are non-renewable (ACSSU116)

- considering what is meant by the term 'renewable' in relation to the Earth's resources
- comparing renewable and non-renewable energy sources, including how they are used in a range of situations

Use and influence of science:
Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations (ACSHE120)

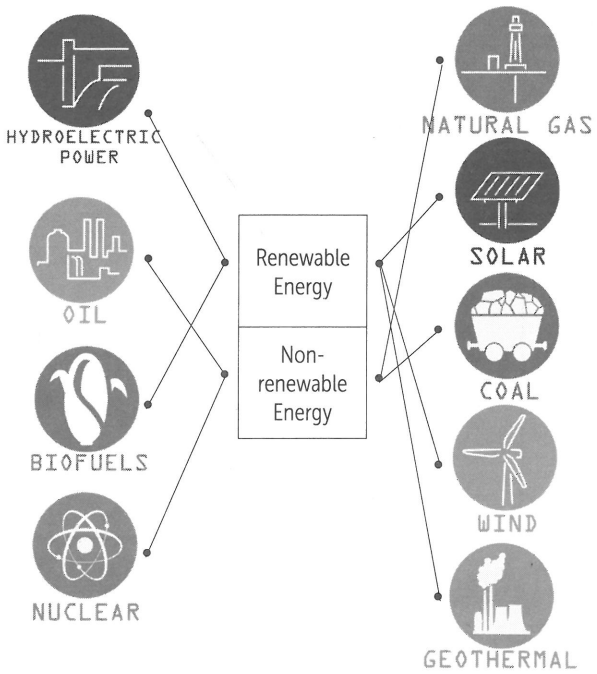
- considering how human activity in the community can have positive and negative effects on the sustainability of ecosystems

LESSON PLAN

Activities	Resources
Activity 1: It's All about Energy! Play Chapters 2 and 3 of the video to introduce the topic. Open the presentation to the first slide and lead a brainstorming session using the following questions: • Have you used electricity today? • When did you use it? • Where does electricity come from? • What can you do to reduce the use of non-renewable energy sources? 15	• ClickView video *Renewable Fuels* Chapter 2: **https://clickv.ie./w/lp/16a** Chapter 3: **https://clickv.ie./w/lp/16b** • Presentation: Renewable and Non-Renewable Energy
Activity 2: Am I Renewable? Use slides 3 and 4 of the presentation to provide a definition of Renewable and Non-Renewable Energy. Give out the All about Energy! worksheet and ask students to complete Part A while watching Chapter 4 of the video. 10	• Presentation: Renewable and Non-Renewable Energy • Photocopies of the All about Energy! worksheet • ClickView video *Renewable Fuels* Chapter 4: **https://clickv.ie./w/lp/16c**
Activity 3: Tell Me About Energy! This is a research activity where students are required to work in pairs. They must choose one form of renewable energy and one form of non-renewable energy to research. Ask students to research their chosen energy sources on the Internet and use the table on All about Energy! Part B to document their findings. Once their findings have been accumulated, they are to create a poster using PowerPoint or similar. Students are to present their posters to the class when they are done. 35	• Photocopies of the All about Energy! worksheet • Presentation: Renewable and Non-Renewable Energy • Access to the Internet • Laptops (if students do not have their own, conduct this lesson in a computer lab)

ANSWERS

All about Energy!
Part A:



Features of renewable energy:	Features of non-renewable energy:
• can be replenished naturally • is usually replenished over a short period of time	• is available in limited quantities • takes a long time, usually millons of years, to be replenished

Part B:
Students' answers may vary.

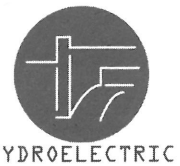


EARTH AND SPACE SCIENCES

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All about Energy!

Part A: Draw a line to match each type of energy to the correct box. Then, use the presentation to help you list two features of each energy category.









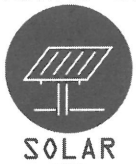
Renewable Energy
Features:

- _____
  _____
- _____
  _____

Non-Renewable Energy
Features:

- _____
  _____
- _____
  _____











Part B: Choose one renewable and one non-renewable energy source to research. Fill in the table below with your notes before transforming your findings into a presentation.

Type of renewable energy:		**Type of non-renewable energy:**	
Where does it come from?		Where does it come from?	
How is it transformed into energy?		How is it transformed into energy?	
Its advantages:	Its disadvantages:	Its advantages:	Its disadvantages:

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